MILITARY SPECIFICATION SHEET

ELECTRON TUBE, THYRATRON

TYPE 5685

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

DESCRIPTION: Triode, xenon gas

See figure 1

Mounting position: Vertical, base down

Weight: 7-ounces nominal (198.5 grams)

ABSOLUTE RATINGS:

Parameter:	Ef	epx	epy	Ib	ib	i surge 0.1 µs	tk	TA
Unit:	V	v	v	Adc	a	a	sec	°C
Maximum:	2.63	1,250	1,000	6.4	77	770	---	75
Minimum:	2.37	---	---	---	---	---	60	-55
TEST CONDITIONS:	2.5	1,250	1,000	---	---	---	---	---

GENERAL:

Qualification - Required

Ⓕ denotes changes

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METHOD	REQUIREMENT OR TEST	NOTES	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
	Quality conformance inspection, part 1								
3201	Critical grid voltage for conduction	-	Ebb = 1,000 Vdc; Rg = 10,000 ohms (max); Rp = 750 to 5,000 ohms	0.65	II	Ecc	-3.0	-6.2	Vdc
(F) 3201	Critical anode voltage for conduction (1)	-	Ecc = 3 Vdc; Rg = 10,000 ohms (max); Rp = 750 to 5,000 ohms	0.65	II	Ebb	---	75	Vdc
(F) 3204	Voltage drop	1,3	Ib = 6.4 Adc; Epp = 130 Vac (max); grid connected to anode through 10,000 ohms	0.65	II	Etd	---	14	Vdc
3206	Operation (1)	1	Rg = 10,000 ohms; Ib = 6.4 Adc; t = 30 seconds (min)	0.65	II	Ecc	-3.0	-6.2	Vdc
3216	Grid current	1,4	Rg = 1 Meg (min) and 10,000 ohms; Ib = 6.4 Adc; t = 15 seconds (min)	0.65	II	Ic	---	10	µAdc
1231	Peak emission by voltage drop	1,5, 8	ib = 77 a; grid connected to anode through 10,000 ohm resistor	0.65	II	etd	---	45	v
(F) 3241	Heater current	-		0.65	II	If	17	23	Aac
	Quality conformance inspection, part 2								
1036	Bump and short	-	Angle = 25°	---	---	---	---	---	---
3201	Critical anode voltage for conduction (2)	1,6	Ecc = 0; Rg = 10,000 ohms (max); Rp = 2,500 ohms	---	---	Epp	---	80	Vac
---	No-load operation	7	Ep = 0; t = 100 hours	---	---	---	---	---	---
	Quality conformance inspection, part 3								
---	Life test	-	Group D; Epp = 220 Vac (min); Rp/Ib = 6.4 Adc; Ecc = 0 to 2 V; Rg = 10,000 ohms (min); t = 500 hours	---	---	---	---	---	---
---	Life-test end points:								
1231	Peak emission by voltage drop	-		---	---	etd	---	67	v
3206	Operation (1)	-		---	---	Ecc	-3.0	-6.2	Vdc
3216	Grid current	-		---	---	Ic	---	10	µAdc
3206	Operation (2)	1,2, 9	Rg = 10,000 ohms; Ib = 6.4 Adc; t = 30 seconds (min); t = 1 hour (min)	---	---	Ecc	-3.0	-6.2	Vdc

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NOTES:

1. Filament pin adjacent to the grid pin shall be negative with respect to other filament pin on the half-cycle that anode is positive.

2. The test is to be conducted in a chamber at a pressure of 54 mmHg. The tube shall not arc or break down externally to the envelope, between the base leads, or fail to operate for any reason.

3. Wattmeter method: With an alternating current source and resistive load, measure the power loss in the tube with a wattmeter so that the current coil is in series with, and the voltage coil across, the tube. The tube voltage drop (Etd) is the total power loss minus the coil loss under load, divided by the average, or dc, current (Ib).

4. Operate tube at least 5 minutes prior to test. Operation may be at optional Epp but Ib shall be 6.4 Adc (min). To test, switch to test voltage and Ib = 6.4 Adc in less than 2 seconds.

5. Adjust Epp and Rp to give required current.

6. Operate tube for 1 minute with Ib = 6.4 Adc before reading critical anode voltage for conduction (2).

7. After no-load operation the tube shall be within specification limits for operation (1) and grid-current tests. In operation (1) test, the operating time shall be a minimum of 30 and a maximum of 120 seconds.

8. This test is to be the first test performed at the conclusion of the holding period.

(F) 9. This test shall be performed annually. A sample size of three tubes shall be used, with no failures allowed. In case of a sample failure, the test shall become a quality conformance inspection (QCI), part 2 test, for three successful consecutive submissions, at which time the test may revert to the QCI, part 3 test.

Custodians:
Army - EL
Navy - EC
Air Force - 85

Review activities:
Air Force - 11, 17, 80
DSA - ES

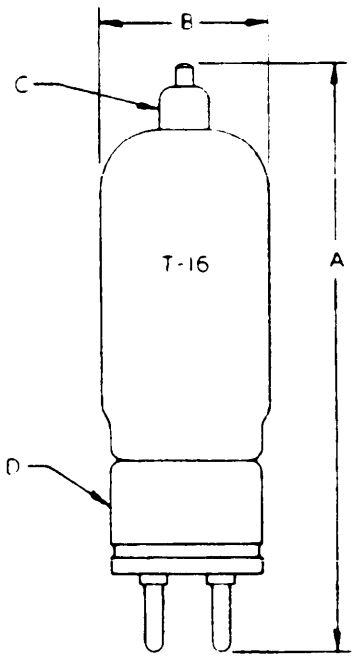
User activities:
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Preparing activity:
Navy - EC

Agent:
DSA - ES

(Project 5960-2883)

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PIN CONNECTIONS	
Pin No.	Element
1	g
2	f
3	f
4	nc
cap	a

Ltr	Dimensions in inches with metric equivalents (mm) in parentheses (see note)	
	Minimum	Maximum
Quality conformance inspection, part 2		
A	8.00 (203.2)	9.50 (241.3)
B	---	2.10 (53.3)
(F) Quality conformance inspection, part 3 (see note)		
C	Cap: C1-5 with ceramic collar (EIA)	
D	Base: A4-81 (EIA)	

NOTE:
A sample size of three tubes shall be checked annually, with no failures allowed. In case of a sample failure, the test shall become a quality conformance inspection (QCI), part 2 test, for three successful consecutive submissions at which time the test may revert to the QCI, part 3 test.

FIGURE 1. Outline drawing of electron tube type 5685.

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WASHINGTON, D. C. 20360

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